

Bounds on the k -Domination Number of a Graph

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Abstract

The k -domination number of a graph is the cardinality of a smallest set of vertices such that every vertex not in the set is adjacent to at least k vertices of the set. We prove two bounds on the k -domination number of a graph, inspired by two conjectures of the computer program Graffiti.pc. In particular, we show that for any graph with minimum degree at least $2k - 1$, the k -domination number is at most the matching number.

1. Introduction

For a positive integer k , a k -dominating set of a graph G is a set S of vertices such that every vertex in $V(G) \setminus S$ has at least k neighbors in S . For a graph G , the minimum cardinality of a k -dominating set is called the k -domination number of G , and is denoted $\gamma_k(G)$. This invariant was introduced by Fink and Jacobson [7], and has been studied by a number of authors including [3, 5, 6, 8, 9, 10, 11].

We will use some standard terminology from graph theory, for which we refer the reader to [1]. The *independence number* of a graph G is the cardinality of an independent set of maximum size, and will be denoted $\alpha(G)$. The *matching number* of a graph G is the cardinality of a matching of maximum size in G , and will be denoted $\alpha'(G)$.

If S is a set of vertices of G , then $G[S]$ will denote the subgraph of G induced by S , and $G - S$ will denote the subgraph of G induced by $V(G) \setminus S$. The degree of a vertex v will be denoted $d(v)$, and the minimum degree of G will be denoted $\delta(G)$.

The following result is due to Caro and Roditty [3]:

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Theorem 1. *Let r and k be positive integers. Let G be a graph of order n where $\delta(G) \geq \frac{r+1}{r}k - 1$. Then*

$$\gamma_k(G) \leq \frac{r}{r+1}n.$$

We will need the $r = 1$ version of this theorem. Namely:

Corollary 2. *Let G be a graph of order n where $\delta(G) \geq 2k - 1$. Then*

$$\gamma_k(G) \leq n/2.$$

In this note, we will improve and generalize Corollary 2. Our first result is the following:

Theorem 3. *Let k be a positive integer, and G a graph of order n . Let $H \subseteq V(G)$ be the set of vertices of degree less than $2k - 1$. Then*

$$\gamma_k(G) \leq \alpha'(G - H) + |H|.$$

If we suppose that H is empty, we get the following succinct result:

Corollary 4. *Let k be a positive integer. For any graph G with $\delta(G) \geq 2k - 1$,*

$$\gamma_k(G) \leq \alpha'(G).$$

To see that equality can be achieved in the corollary above, even for graphs that do not have perfect matchings, consider a complete bipartite graph with $2k - 1$ vertices in one part and more than $2k - 1$ vertices in the other part. Note that the special case $k = 1$ of Corollary 4 is attributed to Hedetniemi in [2].

Our second result is the following:

Theorem 5. *Let k be a positive integer, and G a graph of order n . Suppose that in G no two vertices of degree less than $2k - 2$ are adjacent. Let $H \subseteq V(G)$ be the set of vertices of degree less than $2k - 1$. Then*

$$\gamma_k(G) \leq \frac{n + \alpha(G[H])}{2}.$$

Complete graphs of order $2k - 1$ not only demonstrate that this bound is sharp for every k , but also provide examples where this bound is sharp while Theorem 3 is very weak and Corollary 2 cannot even be applied. Now, if we suppose $k = 2$ and the graph is a connected and bipartite, then since $n \leq 2\alpha(G)$ and $\alpha(G[H]) \leq \alpha(G)$, we get the following result of Fujisawa et al. [8]:

Corollary 6. *If G is a connected bipartite graph of order at least 3, then*

$$\gamma_2(G) \leq \frac{3}{2}\alpha(G).$$

These results were inspired by two conjectures of the computer program Graffiti.pc. The program conjectured the special cases of Theorems 3 and 5 where $k = 2$ (see Conjectures 388 and 392a of [4]), and these conjectures were announced at the Southeastern Conference on Combinatorics, Graph Theory and Computing, held in Boca Raton, March 2010.

2. Proof of Theorem 3

We need the following folklore result:

Lemma 7. *For any graph G , $V(G)$ can be partitioned into two parts S and T , such that each vertex v in S has at least $d(v)/2$ neighbors in T , and each vertex $w \in T$ has at least $d(w)/2$ neighbors in S .*

Proof. Consider the partition of $V(G)$ into S and T such that the number of edges between S and T is maximized. Then any vertex must have at least half its neighbors in the other part. $\square$

Proof of Theorem 3. By Lemma 7, there exists a partition of the vertices of $G - H$ into two parts S and T such that each vertex in S has at least half its neighbors in T , and each vertex in T has at least half its neighbors in S .

Let B be the bipartite subgraph of $G - H$ consisting of the edges that are between S and T , and let M be a maximum matching in B . Let A be the subset of S containing those vertices that are unmatched by M . If $A = \emptyset$, then define $C = D = \emptyset$. Otherwise, consider the set of vertices that are reachable from A by an M -alternating path. Let C be the subset of S that is reachable in this way, and D the subset of T that is reachable in this way. Note that $A \subseteq C$.

By the maximality of M , there is no M -augmenting path in B and so all vertices in D are matched by M . Furthermore, by the construction, M matches each vertex in D with a vertex in C . It follows that

$$|M| = |D \cup (S \setminus C)|.$$

Note that by the construction, there are no edges, in $G - H$, between C and $T \setminus D$. Thus, for any vertex in C , at least half its neighbors are in D . Similarly, for any vertex in $T \setminus D$, at least half its neighbors are in $S \setminus C$.

Let

$$F = D \cup (S \setminus C) \cup H.$$

We claim that F is a k -dominating set for G . For, consider any vertex v that is not in F . As v is not in F , it is not in H either, and so has degree at least $2k - 1$. At least half of the neighbors of v are in F , since any neighbor of v that belongs to H is in F , and at least half of the remaining neighbors are in F . It follows that v has at least k neighbors in F .

Thus,

$$\gamma_k(G) \leq |M| + |H| \leq \alpha'(G - H) + |H|.$$

$\square$

3. Proof of Theorem 5

Proof of Theorem 5. By the assumption, the set of vertices of degree less than $2k - 2$ is an independent set. Let I be a maximal independent set in $G[H]$ containing all vertices of G of degree less than $2k - 2$, and let $J = G - I$. Our

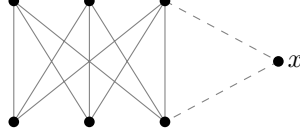


Figure 1: The graph F , for $k = 2$.

strategy will be to construct a k -dominating set of G by taking the union of I and a minimum k -dominating set of a graph obtained by augmenting J in a certain way so that we may appeal to Corollary 2.

We will use the complete bipartite graph $F = K_{2k-1, 2k-1}$ to form a graph J^* with $\delta(J^*) \geq 2k-1$ in the following manner. For each vertex x of J of degree less than $2k-1$: introduce $\lceil ((2k-1) - d_J(x))/2 \rceil$ copies of F and attach each to x by two edges such that the ends of the edges are adjacent. See Figure 1.

Let D^* be a minimum k -dominating set of J^* . We claim that D^* contains at least $2k-1$ vertices from each attached F . For, if D^* has less than k vertices from one partite set of the F , then it must have every vertex from the other partite set except possibly the vertex w attached to x ; and if vertex $w \notin D^*$ then at least $k-1$ vertices from the other partite set must be in D^* .

Also, we claim that we can choose D^* so that it has exactly $2k-1$ vertices from each attached F . For, if it has more, these can be re-arranged to be one partite set and x . Further, by considering all the possibilities, it follows that an x attached to a F has exactly one neighbor in that F that is in D^* .

Since $\delta(J^*) \geq 2k-1$, we know from Corollary 2 that $|D^*| \leq |V(J^*)|/2$. Set $D = I \cup (D^* \cap V(J))$. From the above it follows that

$$|D| \leq |I| + \frac{|V(J)|}{2} = \frac{n + |I|}{2} \leq \frac{n + \alpha(G[H])}{2}. \quad (1)$$

It remains only to show that D is a k -dominating set of G .

Note that all vertices of J that had no F graphs attached are k -dominated by D . So, let x be a vertex which had at least one of the F graph attached. If $x \in D$ then there is no problem; so assume $x \notin D$.

By the choice of I , $d_G(x) \geq 2k-2$. By the definition of J , vertex x has $d_G(x) - d_J(x)$ neighbors in I . If $d_G(x) \geq 2k-1$, then x has at most $\lceil (d_G(x) - d_J(x))/2 \rceil$ neighbors in $D^* \setminus D$. Since x is k -dominated by D^* , it has at least $k - \lceil (d_G(x) - d_J(x))/2 \rceil$ neighbors in J , and therefore at least k neighbors in D . That is, x is k -dominated by D .

So assume $d_G(x) = 2k-2$. Then $d_J(x) < 2k-2$, since otherwise we contradict the maximality of I . It follows again that vertex x has at least as many neighbors in I as it has in $D^* \setminus D$, and so is k -dominated by D .

Consequently, D is a k -dominating set of G , and together with (1), this completes the proof. $\square$

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